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PyStarshade: simulating high-contrast imaging of exoplanets with starshades

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Summary

A key challenge in detecting exoplanets and characterizing their atmospheres is distinguishing an exoplanet's faint reflected or emitted light from that of the star they orbit. Direct-imaging addresses this by suppressing light from the star to reveal orbiting exoplanet's (Traub & Oppenheimer, 2010). Direct imaging can be achieved through two approaches: internal occulters, called coronagraphs, which use optics within the telescope to block starlight, and external occulters, called starshades, petal-shaped masks some 10's of meters across in diameter, flown in tandem with a telescope (Arenberg et al., 2021). Current direct imaging efforts using ground-based coronagraphs like the Very Large Telescope (VLT) and Gemini Observatory have successfully imaged young, massive exoplanets at wide separations (Follette, 2023). However, seeing a faint Earth-like planet in the habitable zone of a Sun-like star will require extreme starlight suppression to the level of 10^8 - 10^{10} star-to-planet flux ratio (Angel et al., 1986).

While a starshade has not yet been flown in space, subscale starshades have demonstrated $\leq 10^9$ starlight suppression at a wavelength of 633 nm in laboratory demonstrations (Harness et al., 2021), and $\leq 10^{10}$ across the broad wavelength range necessary for detecting biomarkers like water, oxygen, and methane in computational simulations (Mennesson et al., 2024). Beyond demonstrating starlight suppression, critical questions remain regarding the properties and orbital configurations of exoplanetary systems observable by starshade missions, as well as optimal instrument designs. `PyStarshade` is a Python toolbox to efficiently perform optical simulations of starshades with exoplanetary scenes to assess the utility of starshades in future direct-imaging missions.

Several starshade concepts have been proposed but not adopted, including a 26 m diameter design to rendezvous with the Nancy Grace Roman Space Telescope (a coronagraph scheduled for launch in the mid 2020's) (Romero-Wolf et al., 2021) and a 52 m design as part of the Habex concept (Gaudi et al., 2020). NASA's Habitable Worlds Observatory (HWO) is in the early design phase with the goal to find and characterize a handful of Earth-like exoplanets (National Academies of Sciences & Medicine, 2023). A 60 m starshade operating over visible to infrared wavelengths (500–1000 nm), and a 35 m UV (250-500 nm) starshade, have been proposed for HWO (Shaklan et al., 2023).

Statement of need

`PyStarshade` provides a toolbox for performing optical simulations from source to focal plane with a configurable starshade telescope design. Complex electric fields are calculated at three planes of propagation (starshade, telescope aperture and focal plane) using Fresnel or Fraunhofer diffraction formula where appropriate. First-order imaging characteristics of a starshade can be determined from analytic relations which depend on the starshades size, telescope aperture size,

wavelength and flight distance. PyStarshade allows for second-order imaging characteristics to be studied, including imaging simulations with a pixelized exoplanetary input scene of integrated flux per pixel and wavelength, varying starshade mask and telescope aperture mask to study throughput and post-processing methods. An example is provided in Figure 1, using spectral- and time-dependent scenes consisting of planets, star and a dust-disk, generated with ExoVista (Stark, 2022). PyStarshade is intended to be flexible and efficient in studying i) exoplanet retrievals, and, ii) instrument design.

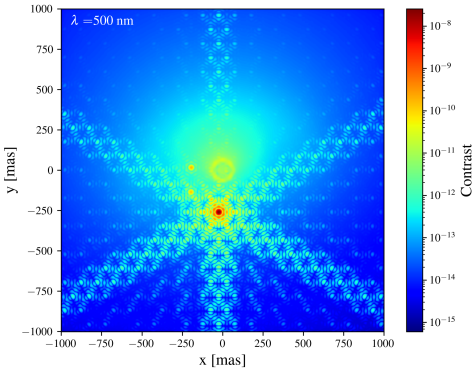

Figure 1: A starshade imaging simulation at a wavelength of 500 nm with a synthetic exoplanetary input scene (generated with ExoVista): three exoplanets are directly visible, while two more sit inside the starshade suppression zone. The scene assumes a 60 m HWO starshade paired with a 6 m segmented telescope; the planets in the scene have planet-to-star flux ratios between 10^{-8} and 10^{-10}.

Fourier optics simulations of starshades and telescope apertures require fine spatial pixel scales to accurately model diffraction without aliasing errors, and across multiple stages of propagation. Traditional Fast Fourier Transforms (FFTs) necessitate large zero-padding factors to accomplish this, increasing computational cost. PyStarshade implements the Bluestein FFT (Bluestein, 1970) (related to the chirp Z-transform) to compute optical fields at arbitrary spatial scales, without zero-padding and therefore with greater efficiency than the FFT or discrete Fourier transform (DFT). In the associated paper submitted to AAS (Taaki et al., 2025), we use PyStarshade to assess the complementary role of a 60 m HWO starshade (Shaklan et al., 2023) paired with two different 6 m segmented and obscured telescope apertures featuring cm-scale details. We report *core throughput* for these apertures, defined as the fraction of exoplanet light recovered. Core throughput is a key metric that governs the exposure time needed to image an exoplanet and therefore the overall exoplanetary yield of a mission.

Furthermore, PyStarshade implements a novel technique to chunk Bluestein FFT calculations that mitigates memory bottlenecks. A set of unit tests are provided to validate the Fourier diffraction tools. PyStarshade pre-computes fields, point-spread-functions and throughput grids for different stages of the optical propagation chain, allowing for flexibility in modifying instrument parameters or telescope aperture masks. PyStarshade optionally interfaces with HCIPy (Por et al., 2018) to generate telescope apertures and the resulting imaging.

Related software

The Starshade Imaging Simulation Toolkit for Exoplanet Reconnaissance (SISTER) (Hildebrandt et al., 2021) implemented in Matlab performs detailed end-to-end simulations of starshade imaging, further including instrument noise sources such as solar glint, and tools for modeling an exoplanetary system and background objects. The optical simulation pipeline of PyStarshade follows a similar approach to that of SISTER, employing a precomputed PSF basis for image simulation. However, SISTER computes diffraction from the starshade to the telescope aperture

using the boundary diffraction wave method (Cady, 2012), performing a 1D integral along the starshade's edge, followed by a DFT to obtain the final PSF basis. Several other codes focus exclusively on calculating the starshade diffraction incident on the telescope aperture; `Fresnaq` (Barnett, 2021b) is a Matlab code to efficiently compute diffraction of a starshade via the non-uniform FFT (Barnett, 2021a). `Diffraq` (Harness, 2021) is a Python implementation of `Fresnaq`, this toolbox also provides an implementation of the boundary diffraction wave method and tools to generate starshade masks (Harness et al., 2018). Starshade masks used in `PyStarshade` are obtained from `SISTER`, or were generated with `Diffraq`. Outside of imaging simulations, the Exoplanet Open-Source Imaging Mission Simulator (`EXOSIMS`) (Savransky et al., 2019) provides methods for orbital simulation of starshades and fuel consumption calculations.

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